R1D4 Robot User Interface & Swarm Robots

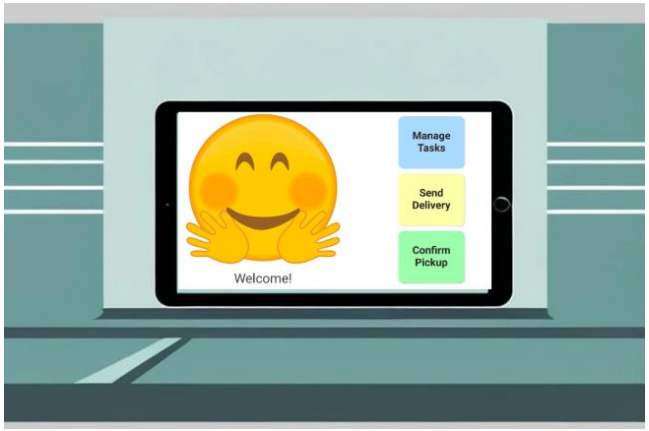
ECSE 458D1: Capstone Design Project
DP 36
Advisor: Joseph Vybihal

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Noorjahan Abonty
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Paolo Bizzarri
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Joseph Hershkop
(260983531)

24th February, 2025

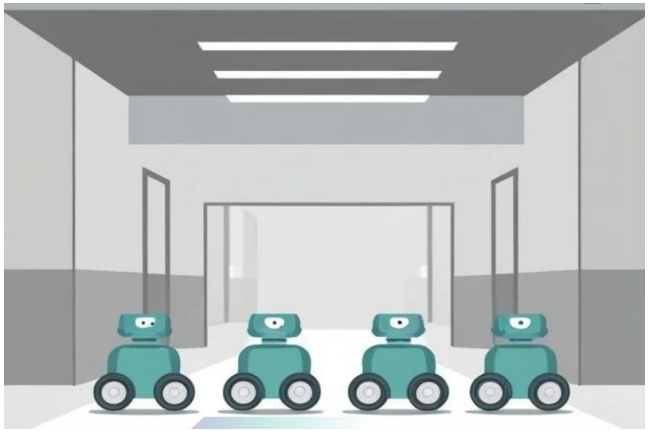
Project Goals

First Semester: R1D4 User Interface



Goal: Intuitive Touch Interface Facial Expressions

Second Semester: Swarm Robots



Goal: Swarm robots can navigate to the exit

Meet Our Team



Project Advisor: Joseph Vybihal

Group: DP 36



Anjali Singhal
261011655



Noorjahan Abonty
260953393



Paolo Bizzarri
261031842



Joseph Hershkop
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R1D4 Robot Interface

What is the R1D4?

- R1D4 is a hospital robot
- Helps hospital staff send deliveries through an App
- Real-time status updates
- Integrates with R1D4 via USB and Telnet server.

Key Success Criteria

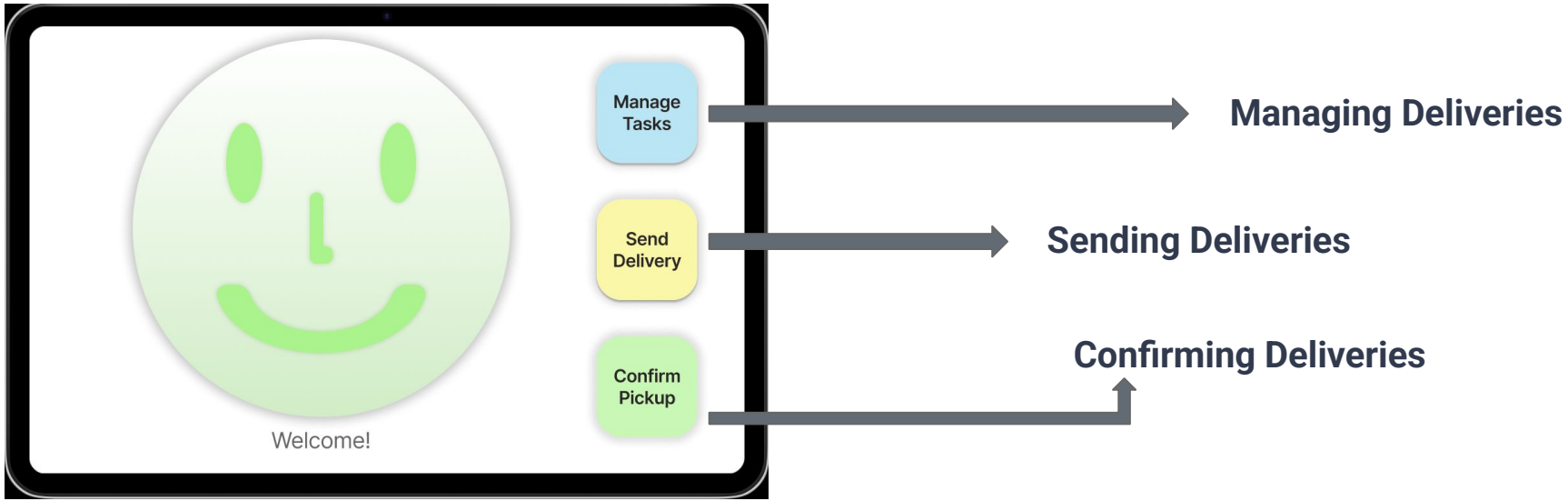
✔ **Expressions: Indicate robot's status and actions**

- **Welcome**
- **Processing**
- **Traveling**
- **Ready**
- **Error**



Key Success Criteria

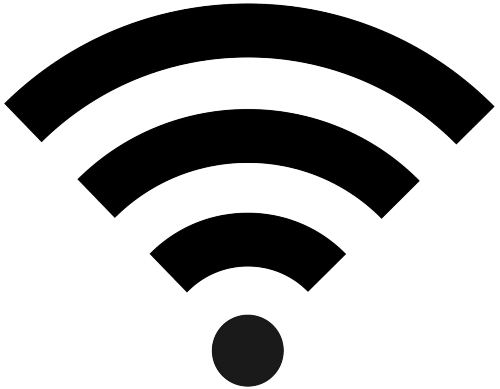
✔ **Delivery Management Features:**

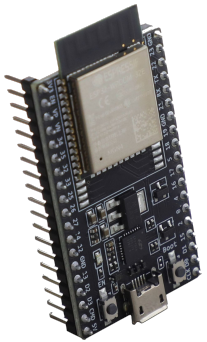


Key Success Criteria

✔ **Reliable Communication Between Components:**







Key Success Criteria

✔ **Intuitive Interface**

- **Considers the hospital staff - minimal training required.**



Final choice

Android Application



Why Android Application Was the Optimal Choice?

🔗 **Reliable Communication**

 **Offline Functionality**

⚡ **Better Performance**

🔒 **Improved Security**

🖥️ **Seamless Hardware Integration**

Assumptions

📍 Accurate Position Tracking

⚠️ Error State Rendering

🔋 Uninterrupted Power Supply

🔗 Stable connection between tablet and ESP32

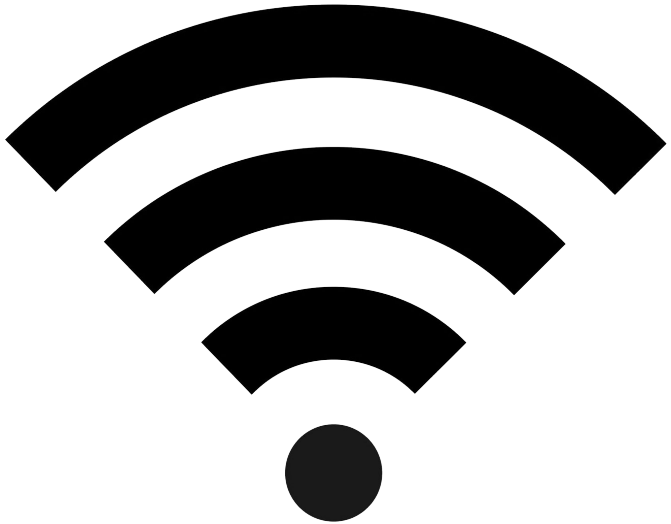
🏥 Predefined Delivery Points

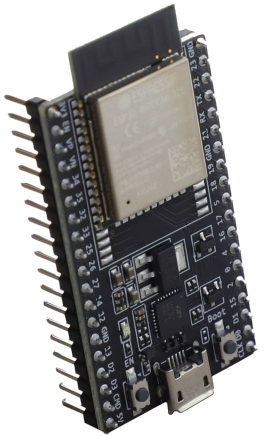
🚧 No Unexpected Obstacles

Communication

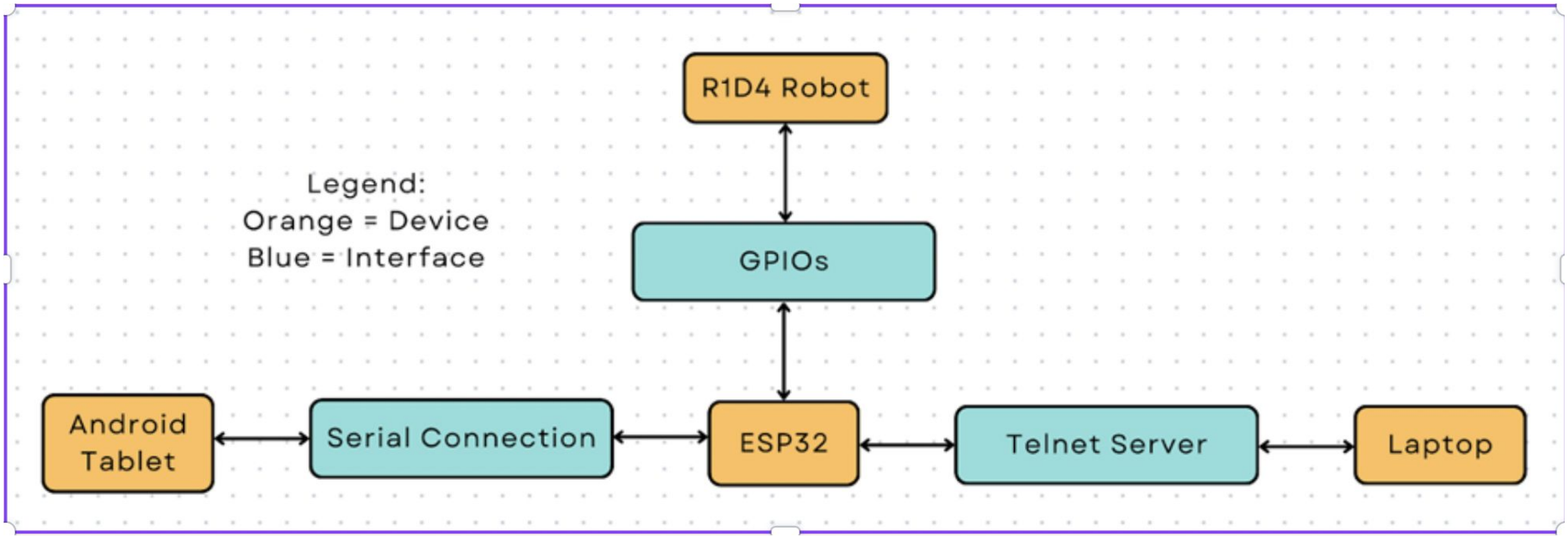
Understanding the Backend





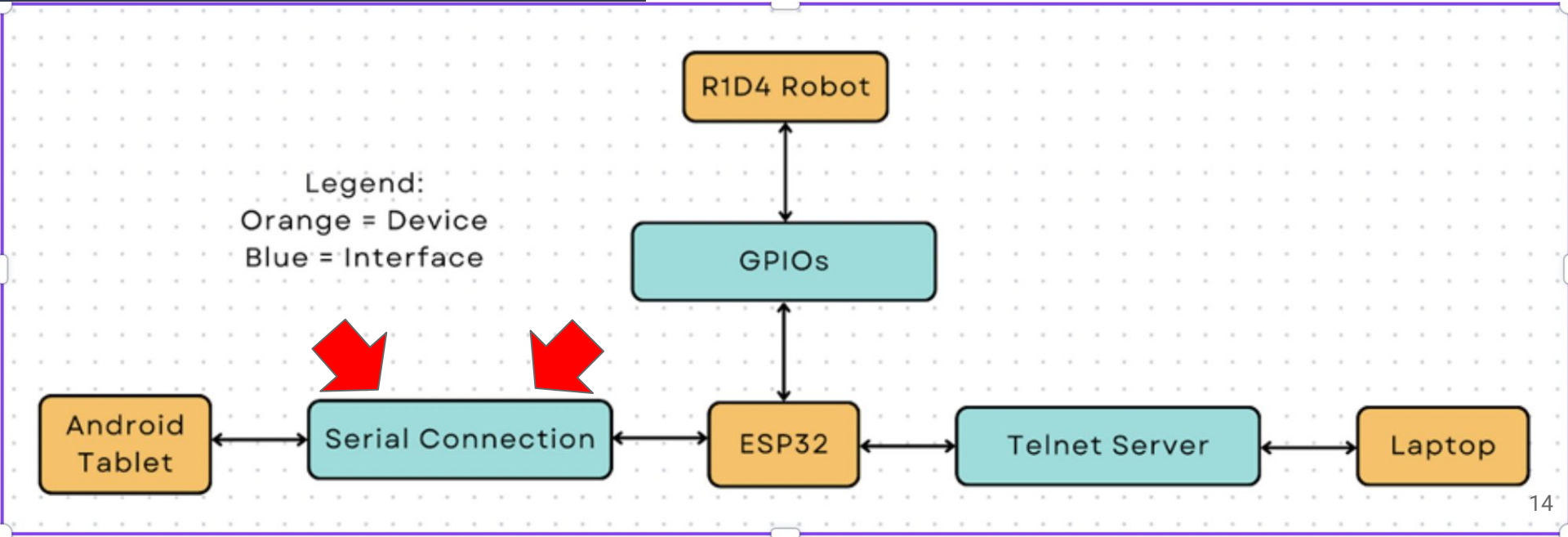


Overall Structure



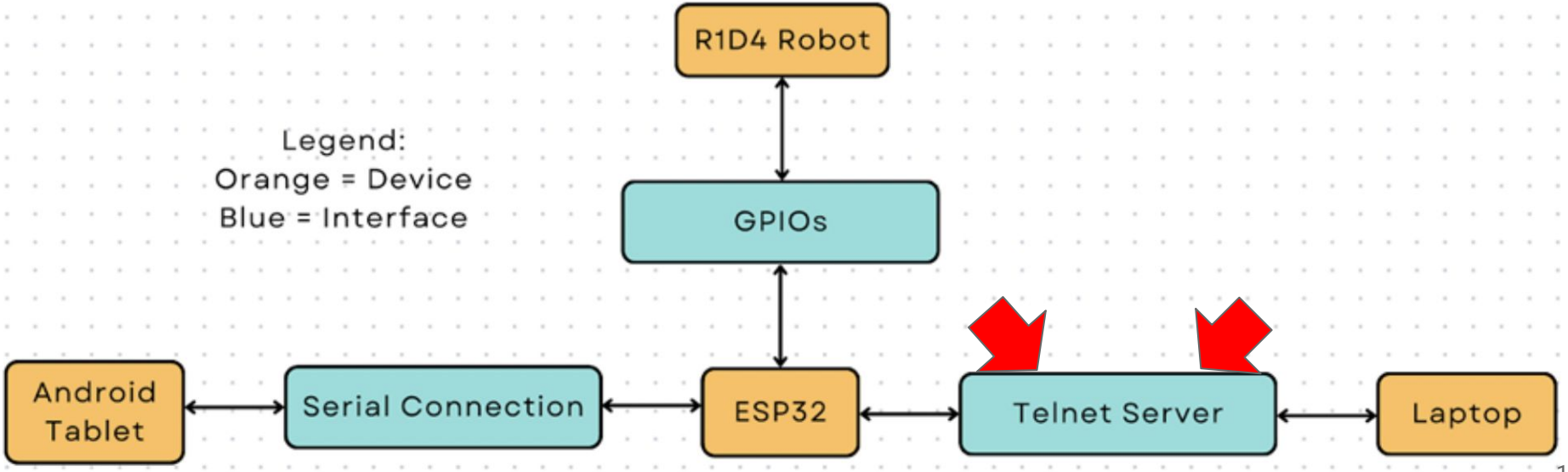
ESP32 ⇔ Tablet

- Serial > Wireless
- Polling
- Permission Required for Connection
- Hex to ASCII Conversion



ESP32 ⇔ Laptop

- Why Telnet?
- Remote User Sends Commands
- Server Runs on ESP32
- Constant Polling



Demo + App Overview

Next Steps - Swarm Bots

HoverBot is our focus

Issues with BoeBots, TankBot

3 Stages:

1. **Initialization**
2. **Navigation (READ)**
3. **Navigation (WRITE)**

HoverBot - Initialization

Configure on-board ESP32 to work with our machine

Get TCP, Middleware servers running

SUCCESS: Manually Send Commands to HoverBot. It Moves.

HoverBot - Navigation (READ)

Python program with live camera feed

- **Device Agnostic**
- **Scalable**
- **No overhead on fragile on-board systems**

Demonstrate object detection (LED, Colour, QR code)

Output “location” based on pixel value

SUCCESS: Location & Orientation of HoverBot

HoverBot - Navigation (WRITE)

UI to choose robot(s) to move, and target location.

- **Visual/Touch based**

Algorithms for orienting HoverBot, moving by X pixels, rotating by X degrees

Incorporating feedback to calculate remaining path

SUCCESS: HoverBot Exits the Room

Paolo - Work Breakdown

ESP32 Initialization

Serial Connection

Proof-Of-Concept Frontend

- **Android App**
  - **Lays good groundwork for the final product**
- **Integrate Serial and Telnet into an Interface**

Next Steps: Navigation (READ)

Anjali - Work Breakdown

Frontend UI

Frontend Delivery Logic

Fixing bugs w/ Face Animations

Parsing delivery info to display over Telnet

Real Time integration and scheduling

Persistent Storage

Next Steps: Navigation (WRITE)

Noor - Work Breakdown

Frontend UI Design

- **Initial Design in Canva**

Mid-Project Report Lead

Project Presentation Lead

- **Structured this presentation**

Next Steps: HoverBot Initialization

Joey - Work Breakdown

Initialize Telnet Server

Send commands to Frontend

- **Trigger face animations**

Swarm Robot Dev Environment Setup

Swarm Robot Navigation planning

Next Steps: HoverBot Initialization

Project Timeline

HoverBot Functioning

- Configure on-board ESP32 to work with our machine
- Get TCP, Middleware servers running
- SUCCESS: Manually Send Commands to HoverBot

Navigation (WRITE)

- Interface to select robot(s) to move and the target location.
- Algorithms for orienting hoverbot, moving by X pixels, rotating by X degrees
- Incorporating feedback to calculate remaining path

Design Day!

- Demo R1D4 Interface
- Demo Swarm Navigation



Navigation (READ)

- Python program with live camera feed
- Demonstrate object detection (LED, Colour, QR code)
- Output “location” based on pixel value

Final Report

- Documentation
- Draft report
- Final report
- Prepare for Design Day
  - Posters

Thank You!